STATEMENT OF WORK

Background and Scope of Services

For more than a decade NASA CMS has supported a growing portfolio of projects shedding light on carbon storage, emissions and removals by coastal and marine systems. Earth observation tools, training materials and scientific outputs are widely deployed by practitioners to advance policy understanding and on-the-ground actions supported by a growing portfolio of market mechanisms. The intent of this project is to review the role of the CMS in supporting these actions.

Blue carbon refers to atmospheric carbon stored in living and non-living plant material and deep sediments of coastal ecosystems, particularly mangroves, salt marshes and seagrass beds. Storage and flows of carbon from coastal wetlands underpins many aspects of the blue economy including fisheries and shoreline stabilization and infrastructure protection leading to additional values that permeate the economy.

When the term blue carbon was introduced over 15 years ago the aim was to raise awareness about limiting and reverting ongoing degradation of Blue Carbon ecosystems (BCEs). Today, the term is widely used in management, scientific, and policy settings alike to reference ecosystem services and sustainable development goals (SDG's) such as climate mitigation and resilient communities (Laffoley & Grimsditch, 2009[1]; Murdiyarso et al., 2015[2]). Blue Carbon ecosystems are found around all continents except Antarctic and across all climate regions.

The assessment will cover the following elements:

- **Activity 1: Summary of the current state and direction of blue carbon science and financing (10 pages)**
    - Review of the latest science for established and emerging blue carbon ecosystems, their carbon sequestration capacity, and their role in climate mitigation
    - Outline the co-benefit opportunities from blue carbon ecosystem conservation and restoration (give examples of successful projects and their impact on local communities and biodiversity)
    - Identify knowledge and data gaps in scaling the science on blue carbon ecosystems
    - Overview of developing finance mechanisms to support emerging blue economies
    - Review of current and emerging standards, methodologies and approaches to blue carbon finance (crediting) mechanisms and opportunities

- **Activity 2: Assessment of the NASA CMS program (10 pages)**
    - Overview of the key components of the CMS program and its main goal of improving monitoring, reporting, and verification (MRV) systems for blue carbon to inform policy and management decision-making
    - Top level survey of how the CMS program has contributed to understanding the global carbon cycle, including blue carbon stocks and fluxes
    - Overview of the remote sensing technologies used by CMS and examples of the type of data products generated from these technologies that are currently being used in blue carbon accounting and monitoring (E.g., aboveground biomass, canopy height, land use land cover change, etc.)

  - Scope what the current CMS project portfolio targets and which datasets are used to inform management and financing.
  - Overview of the CMS program's collaboration and engagement with stakeholders and data users to improve the program.

- **Activity 3: Priorities moving forward for the CMS program to improve blue economy readiness (10 pages)**

  - Review and identify how the CMS program is positioned to fill knowledge gaps in blue carbon science, management and governance (including community governance)
  - Review and identify how CMS data and funding priorities can support implementation of blue economy finance
  - Review and identify how CMS stakeholder engagement processes serve CMS data users in achieving blue economy readiness
  - Provide recommendations for enhancing the CMS program, its contributions to blue carbon monitoring, and its support of overall blue economy readiness

Information will be gathered through targeted interviews with representative CMS project Principal Investigators and end users. A table will be provided that summarizes applications with critical review assessment of whether products meet the needs of end users along with strengths and gaps in the program.

Conclusions will be translated into a short briefing note for senior management and policy makers.

Staffing

The assignment will undertaken by a senior-level team well experienced in translating blue carbon science to policy and market mechanisms.

Deliverables and timeline

Interviews will be undertaken at the NASA CMS meeting in September 2025 and final report provided in Q4 2025